

UniCore Campus Operating Platform

Technical Project Master Report

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Abstract

Modern universities manage student records, hostel allocation, library inventory, and examination processing through disconnected systems, leading to data redundancy, inconsistency, and security risks. This project presents the UniCore Campus Operating Platform, a centralized PostgreSQL relational platform normalized to Boyce-Codd Normal Form (BCNF) across 8 domain schemas and 35+ tables.

1 Problem Statement

Fragmented systems create five critical operational failure modes:

1. **Data Duplication:** Redundant address entries across modules.
2. **Inconsistency:** Mismatched records during partial updates.
3. **Resource Tracking:** Manual tracking of bed capacities and book copies.
4. **Transaction Hazards:** Double-allocation of rooms during peak rushes.
5. **Security Gaps:** Unscoped permissions across modules.

2 Functional Modules & Domain Boundaries

Module	Core Responsibility	Key Entities
Academic	Profiles, CGPA, branching	faculties, departments, courses, students
Hostel	Automated bed allotment	hostels, hostel_rooms, allocations
Library	Inventory, circulation, fines	books, authors, members, borrow_records
Exam	Scheduling, invigilation, grades	exams, exam_schedules, exam_results
Auth	RBAC & sessions	users, roles, permissions, refresh_tokens
Audit	Forensics & mutation logs	audit_logs, login_logs, system_events

Table 1: UniCore Module Boundaries

3 Conclusion

UniCore delivers a reliable, scalable, and secure solution for university administration through strict BCNF normalization, PL/SQL automation, and immutable audit logging.